FRC Research — Weekly Working Notes

GS/MRR Legacy Reproduction, WarpX/ADIOS2 Integration, and Transition to Tilt Dynamics

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1 Weekly objective

The main objective of this week was to close the inherited FRC equilibrium reproduction and move from static GS/MRR initialization to a functioning WarpX tilt-instability simulation.

The work was divided into two stages:

1. reproduce the inherited MRR/Grad–Shafranov equilibrium and seeded $n = 1$ perturbation without modifying the underlying physics;
2. make the inherited BP5 initialization file readable by WarpX and validate a short hybrid-PIC time evolution.

By the end of the week, both stages were completed sufficiently to begin longer tilt simulations.

2 Inherited GS/MRR equilibrium audit

The inherited initializer uses an axisymmetric FRC equilibrium based on the Grad–Shafranov equation,

$$B_z = \frac{1}{r}\frac{\partial \psi}{\partial r}, \qquad B_r = -\frac{1}{r}\frac{\partial \psi}{\partial z},$$

$$\Delta^* \psi = -\mu_0 r^2 \frac{dP}{d\psi}.$$

The MRR-1 pressure closure is

$$\bar{p}(x) = \beta_s e^{-\alpha x}(1 + \sigma x)^n, \qquad n = 2,$$

with normalized flux and pressure

$$x = \frac{\psi}{B_e R_s^2}, \qquad \bar{p} = \frac{P}{B_e^2/(2\mu_0)}.$$

For the inherited case,

$$\beta_s = 0.6, \qquad \delta_s = 0.4\,\delta_{s,\mathrm{RR}},$$

which gives approximately

$$\delta_{s,\mathrm{RR}} \simeq 0.2651, \qquad \delta_s \simeq 0.1060,$$

$$\alpha \simeq 31.5351, \qquad \sigma \simeq 8.31255.$$

The current source code default was found to use

$$B_{\mathrm{ext}} = 0.26 \text{ T}.$$

However, this value did not reproduce the inherited WarpX input.

3 Identification of the legacy magnetic field

A complete initializer run was first performed with the current $B_{\mathrm{ext}} = 0.26$ T default.

The GS solve converged after 408 outer iterations, with

$$I_{\mathrm{target}} \simeq 391.14 \text{ kA}, \qquad I_{\mathrm{mid}} \simeq 391.93 \text{ kA},$$

and final relative change

$$\Delta_{\mathrm{GS}} \simeq 4.999 \times 10^{-5}.$$

Comparison with the inherited WarpX input revealed

$$\frac{n_{0.26}}{n_{\mathrm{legacy}}} \approx 4, \qquad \frac{\psi_{0.26}}{\psi_{\mathrm{legacy}}} \approx 2.$$

These are consistent with

$$P \propto B^2, \qquad n \propto B^2, \qquad \psi \propto B.$$

Since

$$0.26 \text{ T} = 2(0.13 \text{ T}),$$

the observed factors are precisely those expected if the inherited case had originally been generated at

$$\boxed{B_{\text{ext}} = 0.13 \text{ T}}.$$

This interpretation is consistent with the magnetic field value documented in the inherited MRR material.

4 Legacy 0.13 T equilibrium reproduced

The initializer was rerun with

$$B_{\text{ext}} = 0.13 \text{ T}.$$

The reconstructed equilibrium gave

$$I_{\text{target}} \simeq 195.57 \text{ kA}, \qquad I_{\text{mid}} \simeq 195.97 \text{ kA},$$

again converging after 408 iterations to approximately

$$4.999 \times 10^{-5}.$$

The generated WarpX input was then compared directly against the inherited input. Only floating-point differences remained in quantities such as

$$u_0, \qquad R_0, \qquad Z_0, \qquad \psi_O, \qquad N_{\text{tilt}},$$

with relative discrepancies of order

$$10^{-11}$$

or smaller.

> **The inherited 0.13 T seeded MRR/GS initialization was reproduced to numerical precision.**

5 Inherited tilt seed

The inherited case contains a prescribed $n = 1$ axial velocity perturbation. For $\psi < 0$,

$$F(r, \psi) = N \left(\frac{r}{R_0} \right) \left(-\frac{\psi}{\psi_O} \right)^p, \qquad p = 2,$$

with

$$\delta u_z = u_0 F(r,\psi)\cos(\theta-\phi).$$

For the reproduced case,

$$R_0 \simeq 0.1707 \text{ m}, \qquad Z_0 \simeq 0.3244 \text{ m},$$

$$\epsilon = 0.01, \qquad V_A \simeq 1.346\times 10^5 \text{ m s}^{-1},$$

so that

$$\delta u_z^{\max} \simeq 1.346\times 10^3 \text{ m s}^{-1}.$$

An important implementation detail was confirmed: the perturbation is already encoded into the particle momenta written to `init.bp`. Therefore, changing the tilt constants only in the WarpX input does not modify the actual seed. A new `init.bp` must be generated for a true seed-amplitude scan.

6 Initial WarpX failure: BP5 backend missing

A short 200-step WarpX smoke test was prepared using the reproduced 0.13 T initialization.

The existing WarpX executable initialized MPI and OpenMP correctly but terminated before advancing the physical simulation. The openPMD error was

```
openPMD::error::WrongAPIUsage
openPMD-api built without support for backend 'ADIOS2'
```

The previous WarpX CMake cache confirmed

```
ADIOS2_DIR:PATH=ADIOS2_DIR-NOTFOUND
```

Thus the problem was isolated to the I/O/build stack rather than to the equilibrium or hybrid-PIC physics.

7 Local ADIOS2/BP5 build

A separate ADIOS2 v2.11.0 installation was built inside the FRC sandbox so that the existing WarpX installation remained untouched.

The final configuration used approximately

MPI = ON,
SST = OFF,
HDF5 = OFF,
OpenSSL = OFF,
Python = OFF,
Fortran = OFF.

Two SCNet-specific build issues were diagnosed.

7.1 FFS configuration-header collision

The bundled FFS source was resolving the wrong generic `config.h`, producing incorrect compile-time definitions. A build-only workaround was used so that the FFS sources resolved their own generated configuration.

7.2 Unneeded EVPath target

EVPath was still being built even though SST was disabled. A minimal local CMake patch was introduced so that EVPath was only configured when

`ADIOS2_HAVE_SST`

was enabled.

The patch was saved for provenance as

```
adios2-no-evpath-without-sst.patch
SHA256:
04278ae5f6522401d69909b94da68988b324781393794c701deceae70c669996
```

The final ADIOS2 job completed successfully:

Job	124923692
State	COMPLETED
Exit code	0
Elapsed	00:03:45

8 WarpX/openPMD rebuild

A separate WarpX build was prepared from a clean source snapshot while reusing the same local AMReX and PICSAR source trees as the previous working WarpX installation.

The intended chain was

$$\boxed{\text{WarpX} \rightarrow \text{openPMD} \rightarrow \text{ADIOS2/BP5}}.$$

The first rebuild attempt failed during CMake configuration because openPMD attempted to download `nlohmann_json` from GitHub on a compute node without external DNS/network access.

This was not a WarpX or ADIOS2 failure. The previous working WarpX build already contained local copies of the required openPMD dependencies.

The build recipe was corrected to reuse

```
/public/home/acszlf5t66/WQ/warpx-deps/json
/public/home/acszlf5t66/WQ/warpx-deps/toml11
```

instead of attempting network downloads.

The corrected WarpX build then completed:

Job	125246471
State	COMPLETED
Exit code	0
Elapsed	00:14:42
MaxRSS	approximately 6.7 GB

9 Successful 200-step tilt smoke test

The dependent smoke test was then executed automatically using the reproduced 0.13 T GS/MRR equilibrium, the inherited $n = 1$ seed, the same WarpX input physics, and a command-line override of only

$$N_{\text{step}} = 200.$$

The job completed successfully:

Job	125246472
State	COMPLETED
Exit code	0
Elapsed	00:06:53
MPI ranks	4
OpenMP threads/rank	8

WarpX executed all 200 steps,

$$N_{\text{calls}}\,(\texttt{WarpX::Evolve::step}) = 200.$$

The run wrote openPMD diagnostics at

$$n = 0, \qquad 100, \qquad 200,$$

and checkpoints at

$$n = 0, \qquad 200.$$

The global WarpX warning report stated

```
No recorded warnings.
```

at both the first step and the end of the simulation.

> **The WarpX + openPMD + ADIOS2/BP5 integration is now operational, and the inherited seeded FRC case advances successfully in time.**

10 Performance observation

The 200-step evolution required approximately

$$393 \text{ s}$$

inside `WarpX::Evolve()`, corresponding to roughly

$$1.97 \text{ s/step}$$

for this 4-MPI-rank, 8-thread/rank configuration.

The largest runtime contribution was the hybrid-PIC field evolution, especially the repeated electric-field solve.

This should be treated only as an initial reference, not as an optimized production benchmark.

11 Planned tilt diagnostic

The first quantitative instability diagnostic will be based on the complex $n = 1$ axial displacement moment,

$$Z_1(t) = \frac{\int z\, \rho(\mathbf{x}, t) e^{-i\theta}\, dV}{\int \rho(\mathbf{x}, t)\, dV},$$

with amplitude

$$A_1(t) = |Z_1(t)|.$$

During a linear growth interval,

$$A_1(t) \propto e^{\gamma t},$$

so that

$$\ln A_1(t) = \gamma t + C$$

can be used to estimate the physical growth rate.

A normalized quantity such as

$$\gamma \tau_A$$

should not yet be reported until the exact definition of the relevant Alfvén time for this inherited case is identified and documented.

12 New research direction from Prof. Ren

Prof. Ren proposed expanding the long-term goal beyond reproducing the inherited tilt case.

The updated research direction remains centered on suppression of the FRC tilt-mode instability, but should move toward increasingly realistic device design questions, including:

- realistic device radius and length;
- suitable FRC radius, length, and elongation;
- dependence of tilt stability on device geometry;
- formation of the FRC inside an actual device;
- collision and merging of FRCs following a Helion-like sequence;
- subsequent compression;
- identification of physically justified device dimensions;
- eventual integrated simulation/visualization of formation → collision/merging → compression.

The integrated animation is a long-term aspirational deliverable rather than an immediate requirement.

The central engineering question is becoming:

What device dimensions and plasma parameters should be chosen, and what simulation/modeling evidence justifies those choices?

13 Updated framework interpretation

The framework objective should therefore evolve from a reproduction-oriented workflow into a modular FRC simulation and design environment.

A useful high-level sequence is

$$\boxed{\text{Formation} \rightarrow \text{Equilibrium} \rightarrow \text{Tilt stability} \rightarrow \text{Translation/collision} \rightarrow \text{Merging} \rightarrow \text{Compression}}.$$

The current work occupies the equilibrium/tilt part of this sequence.

14 Immediate roadmap

The next work should proceed in the following order.

1. Short tilt evolution

Run a longer but still controlled simulation using the reproduced legacy case. The purpose is to determine when the seeded $n = 1$ perturbation becomes measurably visible.

2. Build the $n = 1$ diagnostic

Implement $A_1(t)$ from the openPMD density diagnostics and inspect its linear-growth regime.

3. Reproduce the inherited tilt behavior

Before modifying geometry or physics, establish a reference growth curve and visual evolution for the inherited case.

4. Directed literature review

Once the baseline tilt behavior is established, perform a targeted literature review focused on device-design quantities such as

$$R_s, \quad L_s, \quad E, \quad R_w/R_s, \quad S^*, \quad B_{\text{ext}}, \quad \text{mirror geometry}.$$

5. Parameterized design studies

Only after the physically meaningful design space is identified should systematic parameter scans begin.

15 Team coordination

A larger FRC project meeting is scheduled for 21:00 on 23 October to discuss division of work among the expanded team.

A natural ownership area for the present work is:

- inherited GS/MRR equilibrium reproduction;
- WarpX/openPMD/ADIOS2 simulation integration;
- baseline tilt simulation;
- $n = 1$ instability diagnostics;
- common simulation-framework/provenance structure.

Other modules such as formation optimization, collision/merging, compression, literature/device constraints, or geometry optimization can be distributed depending on the interests and expertise of the group.

16 Status at the end of 23 October

Task	Status
Formation baseline reproduction	**DONE**
4 μs formation extension	**DONE**
Inherited MRR model audit	**DONE**
Legacy 0.13 T GS equilibrium reproduction	**DONE**
Inherited $n = 1$ seed reproduction	**DONE**
Local ADIOS2/BP5 installation	**DONE**
WarpX/openPMD rebuild with ADIOS2 support	**DONE**
200-step hybrid-PIC tilt smoke test	**DONE**
openPMD diagnostics at steps 0/100/200	**DONE**
Longer tilt evolution	**NEXT**
Quantitative $n = 1$ growth diagnostic	**NEXT**
Directed device-design literature review	**PENDING**
Geometry/tilt parameter scans	**PENDING**
Collision/merging module	**PENDING**
Compression module	**PENDING**
Integrated device-scale sequence	**PENDING**

17 Main weekly conclusions

1. The inherited FRC initializer was identified as a $B_{\text{ext}} = 0.13$ T case despite the newer source default of 0.26 T.
2. The 0.13 T MRR/GS equilibrium and seeded $n = 1$ perturbation were reproduced to numerical precision.
3. The original WarpX executable could not read the BP5 initialization because its openPMD build lacked ADIOS2 support.
4. A local MPI-enabled ADIOS2/BP5 installation was built successfully.
5. A corrected isolated WarpX/openPMD build was completed using local dependencies and the new ADIOS2 installation.
6. The resulting executable successfully evolved the inherited hybrid-PIC case for 200 steps, produced diagnostics and checkpoints, and reported no recorded WarpX warnings.
7. The infrastructure blocker is closed. The next phase is the actual physical characterization of the tilt instability.
8. The long-term project scope has expanded toward realistic device dimensions, tilt suppression, collision/merging, compression, and simulation-informed engineering design.

GS/MRR reproduced → WarpX/BP5 operational → Tilt dynamics → device design

18 Team structure and proposed research role

The broader project is now being organized as a collaborative group of approximately six to seven students under the working theme

FRC Simulation and Design.

Given the expanded scope, the project should not be approached as a single simulation task. A more useful organization is to divide the complete FRC device problem into connected physical and numerical modules, for example:

- FRC formation and pre-ionization;
- equilibrium reconstruction and initialization;
- tilt-mode stability and suppression;
- translation, collision, and merging;
- magnetic compression;
- device, coil, and chamber geometry;
- diagnostics, validation, visualization, and common simulation infrastructure.

The different modules should eventually converge toward a common integrated device model,

$$\boxed{\text{Formation} \rightarrow \text{Equilibrium/Stability} \rightarrow \text{Translation} \rightarrow \text{Collision/Merging} \rightarrow \text{Compression}}.$$

Proposed personal focus

A particularly suitable research direction for the present work is the interface between

$$\boxed{\text{FRC equilibrium and tilt stability} \quad + \quad \text{realistic device geometry/design}}.$$

The immediate work would continue from the reproduced GS/MRR equilibrium and the operational WarpX hybrid-PIC tilt case, but the longer-term objective would be to determine how physically meaningful geometric and plasma parameters affect FRC stability.

Candidate design variables include

$$R_s, \qquad L_s, \qquad E = \frac{L_s}{2R_s}, \qquad R_w, \qquad Z_w, \qquad \frac{R_w}{R_s}, \qquad B_{\text{ext}},$$

together with parameters such as mirror geometry and S^*.

The central scientific question can then evolve from simply

"Does the FRC tilt?"

toward

How do FRC and device dimensions affect tilt stability, and what geometry provides a physically realistic design

This direction connects plasma physics, numerical simulation, instability analysis, and engineering design. It also provides a natural interface with other members working on formation, merging, and compression.

Role in the common framework

The simulation framework can remain a shared contribution, particularly for

- common WarpX configuration;
- run provenance and reproducibility;
- consistent input/output formats;
- diagnostics and visualization;
- transfer of plasma states between different simulation modules.

However, framework maintenance should support the scientific work rather than become the primary research objective.

The preferred personal contribution is therefore to use the framework to produce physically interpretable results on equilibrium, tilt stability, and device geometry, while maintaining enough integration knowledge to connect these results with the formation, merging, and compression work of the rest of the team.

Coordination objective

During the group meeting, an important organizational goal is to define clear interfaces between modules. Each stage should specify both what it receives and what it provides to the next stage.

For example,

$$\text{Formation} \rightarrow \{n,\ T_i,\ T_e,\ \mathbf{B},\ \mathbf{u},\ \psi,\ \text{geometry}\} \rightarrow \text{Merging/Compression}.$$

A common parameter definition and output convention would make it possible to combine independently developed modules later into a complete FRC formation–collision–compression simulation.

This division would allow individual team members to develop scientifically distinct problems while still contributing to the common long-term objective of determining realistic and defensible dimensions for an experimental FRC device.

19 Two-microsecond tilt baseline and first $n = 1$ diagnostic

After validating the 200-step WarpX smoke test, the reproduced $B_{\text{ext}} = 0.13$ T seeded FRC case was extended to 2000 time steps without modifying the underlying physics or initialization.

The production run used

$$\Delta t = 10^{-9} \text{ s}, \qquad N_{\text{step}} = 2000,$$

corresponding to a total simulated time of

$$\boxed{t_{\text{final}} = 2.0 \ \mu\text{s}}.$$

The job completed successfully:

Job	125344674
State	COMPLETED
Exit code	0
Elapsed wall time	01:05:50
Allocated CPUs	32
Peak memory	approximately 7.25 GB
Output size	approximately 3.3 GB

WarpX reported no recorded warnings at either the first step or the end of the run.

19.1 openPMD diagnostic series

The run produced openPMD/BP5 diagnostics every 100 simulation steps,

$$n = 0,\ 100,\ 200,\ \ldots,\ 2000,$$

giving a total of

$$\boxed{\text{21 diagnostic snapshots}}.$$

Since

$$100\Delta t = 0.1 \ \mu\text{s},$$

the diagnostic series provides a time resolution of

$$\Delta t_{\text{diag}} = 0.1 \ \mu\text{s}.$$

The available mesh quantities are

$$\mathbf{B}, \qquad \mathbf{j}, \qquad \rho_{\text{protons}}.$$

No individual particle data are written in this diagnostic series.

The mesh shape is

$$(N_z, N_y, N_x) = (208, 64, 64),$$

with openPMD axis ordering

$$(z, y, x).$$

Both ρ_{protons} and the magnetic-field components are cell-centered, with relative mesh positions

$$\mathbf{x}_{\text{position}} = (0.5, 0.5, 0.5).$$

The spatial resolutions are approximately

$$\Delta z = 1.333 \times 10^{-2} \text{ m},$$

$$\Delta x = \Delta y = 1.155 \times 10^{-2} \text{ m}.$$

19.2 Basic state evolution

A first comparison of the initial and final states showed that the overall plasma configuration remains well behaved over the simulated interval.

At $t = 0$,

$$\begin{aligned} \rho_{\max} &\simeq 68.02, \\ B_{z,\min} &\simeq -0.1124 \text{ T}, \\ B_{z,\max} &\simeq 0.2654 \text{ T}, \end{aligned}$$

while at $t = 2$ μs,

$$\begin{aligned} \rho_{\max} &\simeq 66.12, \\ B_{z,\min} &\simeq -0.1105 \text{ T}, \\ B_{z,\max} &\simeq 0.2620 \text{ T}. \end{aligned}$$

The persistence of negative B_z values indicates that field reversal remains present over this interval.

The integrated proton charge changes only slightly. At the final diagnostic,

$$\frac{Q(t = 2\ \mu\text{s})}{Q_0} = 0.99631,$$

corresponding to an integrated charge reduction of approximately

$$\boxed{0.37\%}$$

over the first 2 μs.

This provides no indication of catastrophic particle loss or numerical collapse during the baseline run.

19.3 First quantitative $n = 1$ tilt diagnostic

To quantify the seeded tilt perturbation, a complex $n = 1$ axial displacement moment was constructed from the proton charge-density field:

$$Z_1(t) = \frac{\int z\, \rho_{\mathrm{protons}}(\mathbf{x}, t) e^{-i\theta}\, dV}{\int \rho_{\mathrm{protons}}(\mathbf{x}, t)\, dV},$$

where

$$\theta = \mathrm{atan2}(y, x).$$

The corresponding mode amplitude is

$$A_1(t) = |Z_1(t)|.$$

Two versions of the diagnostic were evaluated:

1. using all cells with positive proton charge density;
2. using only the denser FRC region,

$$\rho_{\mathrm{protons}} \geq 0.05\, \rho_{\max}(t).$$

The agreement in scale between these two definitions provides a first robustness check against sensitivity to low-density regions.

Using the reproduced reference radius

$$R_0 = 0.1706747 \text{ m},$$

the full-density diagnostic evolves approximately as

$$\begin{aligned} A_1(0) &\simeq 0.0033 \text{ mm}, \\ A_1(0.5\ \mu\text{s}) &\simeq 0.090 \text{ mm}, \\ A_1(1.0\ \mu\text{s}) &\simeq 0.134 \text{ mm}. \end{aligned}$$

The maximum observed value is approximately

$$\boxed{A_1^{\max} \simeq 0.141 \text{ mm}}$$

near

$$t \simeq 1.1\ \mu\text{s}.$$

After this initial rise, the amplitude remains approximately in the range

$$A_1 \simeq 0.126\text{–}0.131 \text{ mm}$$

through the latter part of the simulation.

The normalized displacement remains small,

$$\frac{A_1}{R_0} \sim 7.5 \times 10^{-4},$$

during the late part of the present run.

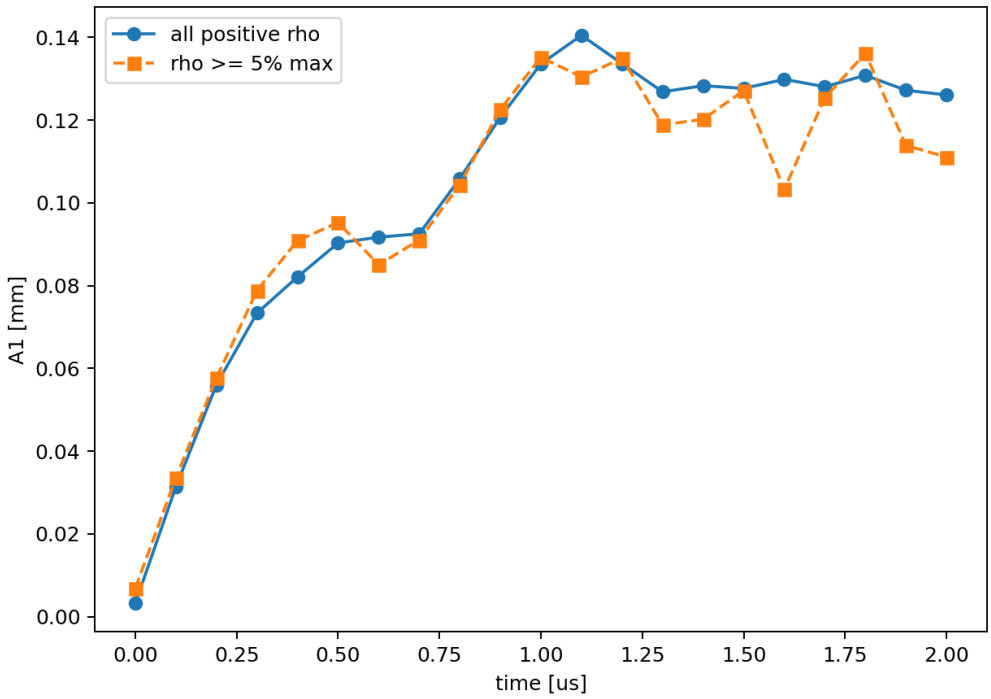


Figure 1: Evolution of the $n = 1$ displacement amplitude $A_1(t)$ during the first 2 μs. The full-density diagnostic and the $\rho \geq 0.05\rho_{\max}$ core diagnostic remain on comparable scales.

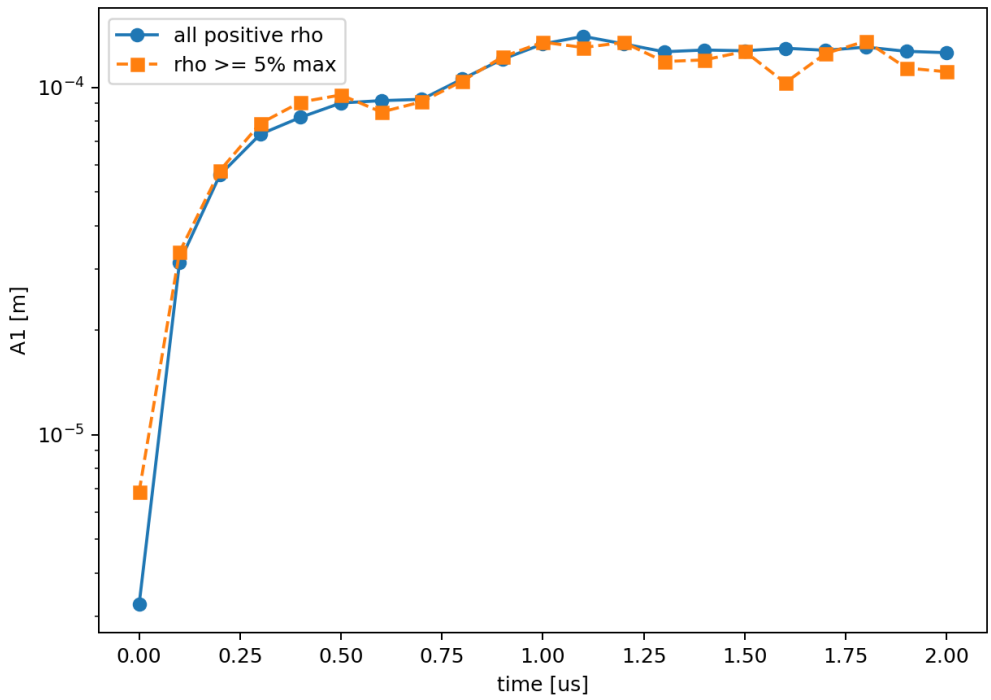


Figure 2: Semilogarithmic representation of the $n = 1$ displacement amplitude. A sustained interval of exponential growth is not yet clearly visible over the present 0–2 μs window.

19.4 Density evolution

For a first visual inspection of the three-dimensional evolution, x–z density cuts were extracted at

$$t = 0,\ 0.5,\ 1.0,\ 1.5,\ 2.0\ \mu\text{s}.$$

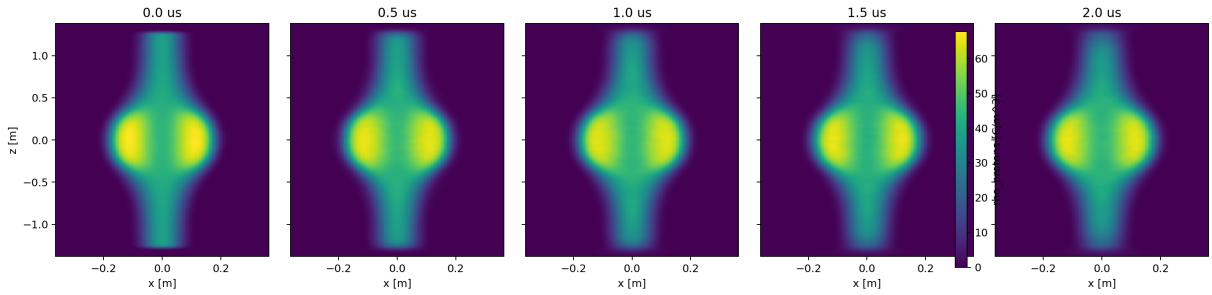


Figure 3: Proton charge-density evolution in an x–z cut for the reproduced seeded FRC baseline. The snapshots correspond to $t = 0$, 0.5, 1.0, 1.5, and 2.0 μs.

19.5 Preliminary physical interpretation

The first 2 μs do not yet show a clear sustained exponential growth regime of the $n = 1$ displacement.

Instead, the present diagnostic shows

Initial response
↓
Growth to $A_1 \sim 0.14$ mm
↓
Approximately constant amplitude

At this stage, the early increase should therefore not automatically be identified with the linear growth rate of the physical tilt instability.

In particular, fitting the complete interval to

$$A_1(t) \propto e^{\gamma t}$$

would not yet provide a defensible value of γ.

The observed behavior may represent an initial response to the imposed velocity perturbation, early plasma relaxation, the beginning of a slower instability, or a combination of these effects. A longer time window and an independent geometric diagnostic are required before distinguishing these possibilities.

Therefore,

γ_{tilt} has not yet been extracted.

19.6 Next diagnostic: FRC centerline tilt

A second, independent diagnostic will be constructed from the transverse center of charge as a function of axial position:

$$x_c(z) = \frac{\int x\rho\, dx\, dy}{\int \rho\, dx\, dy},$$

$$y_c(z) = \frac{\int y\rho\, dx\, dy}{\int \rho\, dx\, dy}.$$

For an approximately rigid tilt, the FRC centerline should satisfy

$$x_c(z) \simeq a_x z + b_x, \qquad y_c(z) \simeq a_y z + b_y.$$

A geometric tilt angle can then be estimated as

$$\theta_{\text{tilt}} \simeq \tan^{-1}\left(\sqrt{a_x^2 + a_y^2}\right).$$

This will provide a physically intuitive measurement that can be compared against the Fourier-like A_1 diagnostic.

19.7 Current conclusion and next simulation decision

The first production tilt baseline has passed the initial numerical validation checks:

2000 steps
2 μs
21 openPMD snapshots
No WarpX warnings
$< 0.4\%$ charge loss

The current interval is sufficient to characterize the initial seeded response but is not yet sufficient to identify a clean exponential tilt-growth phase.

The immediate next steps are therefore:

1. Validate the $n = 1$ result using the centerline-tilt diagnostic.
2. Inspect the density evolution visually.
3. If both diagnostics confirm that the system remains in an early transient or plateau at 2 μs, extend the baseline to approximately
$$t \simeq 5\ \mu\text{s}.$$

4. Use the existing $t = 2$ μs checkpoint for continuation rather than recomputing the first 2000 steps.

5. Extract a tilt growth rate only after a physically defensible linear interval in $\ln A_1(t)$ is identified.

The generated analysis products are

```
img/analysis_n1/A1_vs_time.png
img/analysis_n1/A1_semilog.png
img/analysis_n1/rho_xz_evolution.png
img/analysis_n1/tilt_n1.csv
```

This marks the transition from infrastructure validation to quantitative characterization of the inherited FRC tilt baseline.